

The Sun and Solar System

Advanced Solar Physics and Planetary Science

Cosmic Compass

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Introduction to the Solar System

The solar system consists of the Sun and all celestial objects bound to it by gravity, including planets, moons, asteroids, and comets.

- The Sun is a G-type main-sequence star (G2V).
- It contains 99.86% of the solar system's mass.
- The solar system formed approximately 4.6 billion years ago.

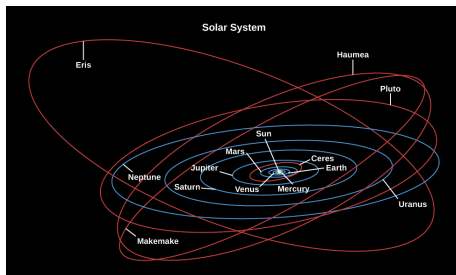


Figure: The Sun and its planets.

Solar Physics: The Sun's Energy

The Sun's energy is produced through nuclear fusion in its core, where hydrogen is converted into helium.

- **Proton-Proton Chain:** The primary fusion process in the Sun.
- **Energy Output:** 3.8×10^{26} watts.
- **Solar Structure:** Core, radiative zone, convective zone, photosphere, chromosphere, and corona.



Planetary Science: Comparative Planetology

The solar system's planets can be divided into terrestrial (rocky) and Jovian (gas giants) planets.

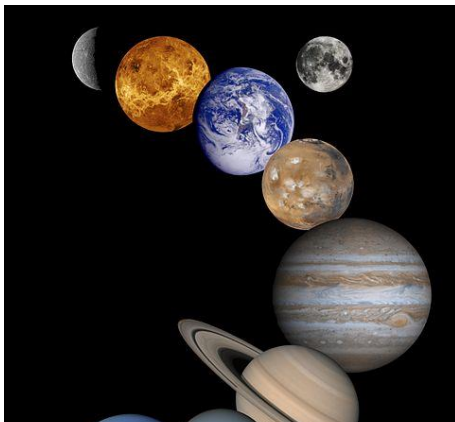
Planet	Distance from Sun (AU)	Diameter (km)	Density (g/cm ³)
Mercury	0.39	4,878	5.4
Venus	0.72	12,120	5.2
Earth	1.00	12,756	5.5
Mars	1.52	6,787	3.9
Jupiter	5.20	142,984	1.3
Saturn	9.54	120,536	0.7
Uranus	19.18	51,118	1.3
Neptune	30.06	49,660	1.6

Table: Comparative data of solar system planets.

Current NASA Missions

NASA's ongoing missions provide valuable data about the Sun and solar system.

- **Parker Solar Probe:** Studying the Sun's corona and solar wind.
- **James Webb Space Telescope:** Observing planetary atmospheres.
- **Artemis Program:** Returning humans to the Moon.



NGSS-Aligned Discussion Questions

- How does the Sun's energy production compare to other stars in the universe?
- What evidence supports the theory of solar system formation?
- How do the properties of terrestrial and Jovian planets differ, and why?
- What are the implications of NASA's missions for future space exploration?

Conclusion

The Sun and solar system are a dynamic and complex system, offering endless opportunities for exploration and discovery.

- Advanced solar physics helps us understand stellar processes.
- Planetary science reveals the diversity of worlds.
- NASA missions continue to expand our knowledge.

